



Spin-orbit induced charge transfer in $H^+/D^+/T^+-N(^4S)$ collisions

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ABSTRACT

The charge transfer in a nonadiabatic spin-orbit induced process $N(^4S) + H^+ \rightarrow N^+(^3P) + H(^2S)$ is investigated. The cross section and the rate constant are reported and compared with other calculations, correcting the previous values based on approximate potentials.

The calculation is extended to collisions involving deuterium and tritium, which are relevant for the transport codes (such as SOLPS-ITER) used in the modeling of fusion plasma.

The method used is a combination of the Landau-Zener theory with a novel methodology based on one-dimensional Gaussian wavepacket propagation.

1. Introduction

The charge transfer in $N(^4S) + H^+$ collisions has been investigated for applications in astrophysics, such as the studies of the heliosphere [1,2] and molecular clouds [3]. It is also relevant for fusion plasmas where impurity seeding is performed. For fusion plasma, it is essential to include deuterium and tritium ions; nitrogen seeding for ITER plasma was investigated with the SOLPS-ITER code in Ref. [4], taking into account the effects of elastic collisions and charge transfer. In general, the need for charge transfer data in plasma transport codes is recognized [5].

The charge transfer in $N(^4S) + H^+$ collisions can be induced by spin-orbit coupling, rotational coupling, and radial coupling; the relevant potential energy curves (PECs) can be inspected in [6]. The rate constant of the spin-orbit induced charge transfer has been investigated with Landau-Zener theory using only approximate potentials thirty years ago [7]. No further studies have been performed, in particular with *ab initio* PECs.

A more recent study [8] reports values based on the radial and rotational coupling mechanism and compares them to other data. It is concluded that the spin-orbit mechanism is crucial below 20,000 K, and at higher temperatures it can be neglected.

The magnitude of coupling and the closeness of PECs suggest that the short-range region may significantly influence rates of nonadiabatic processes, especially at higher energies.

The cross sections and rate constants of charge transfer processes were studied with the wavepacket propagation method at various energy ranges, with explicit treatment of angular momentum, for $C^{4+}-H$ collisions [9], $Si^{4+}-He$ collisions [10], $H-He^+$ collisions [11]. Those

studies were focused on lower energies than the present research and have not investigated the spin-orbit coupling mechanism.

One-dimensional wavepacket dynamics was used for charge transfer in $S^{3+}-H$ collisions at energies of 1–10 eV (below 0.4 hartree) [12]; however, neither the cross section nor the rate constant was calculated. The follow-up study of two-dimensional dynamics suggests that a one-dimensional approach is reasonable [13].

The isotope effect in charge transfer and possible scaling was studied in H^+-Be collisions with approximate PECs and coupling [14] and in hydrogen collisions with many highly charged ions using Landau-Zener theory (the effect is suggested to be significant even at 1 keV/u in some cases) [15]. Comparison of rate constants in the case of $Si^{2+}-H/D$ charge transfer (with radial coupling as a mechanism) confirms that the known approximate scaling generates significant error below 50,000 K [16]. The isotope effect in $H-He^+$ collisions was found to be small and quickly vanishes with increasing energy [11].

In addition to the good quality of Landau-Zener rate constants, especially at higher temperatures [17], the semiclassical molecular orbital close coupling method was found to be in good agreement with quantum cross sections [18,19]. All of these studies suggest that full quantum treatment is not necessary if higher energies are of interest.

This study proposes a novel methodology in which the transition probability for any impact parameter and energy is estimated from one-dimensional (i.e., at zero impact parameter) Gaussian wavepacket propagation in analogy to the Landau-Zener treatment of three-dimensional scattering. The cross sections and rate constants of charge transfer are reported, and the influence of the short interatomic distances is investigated.

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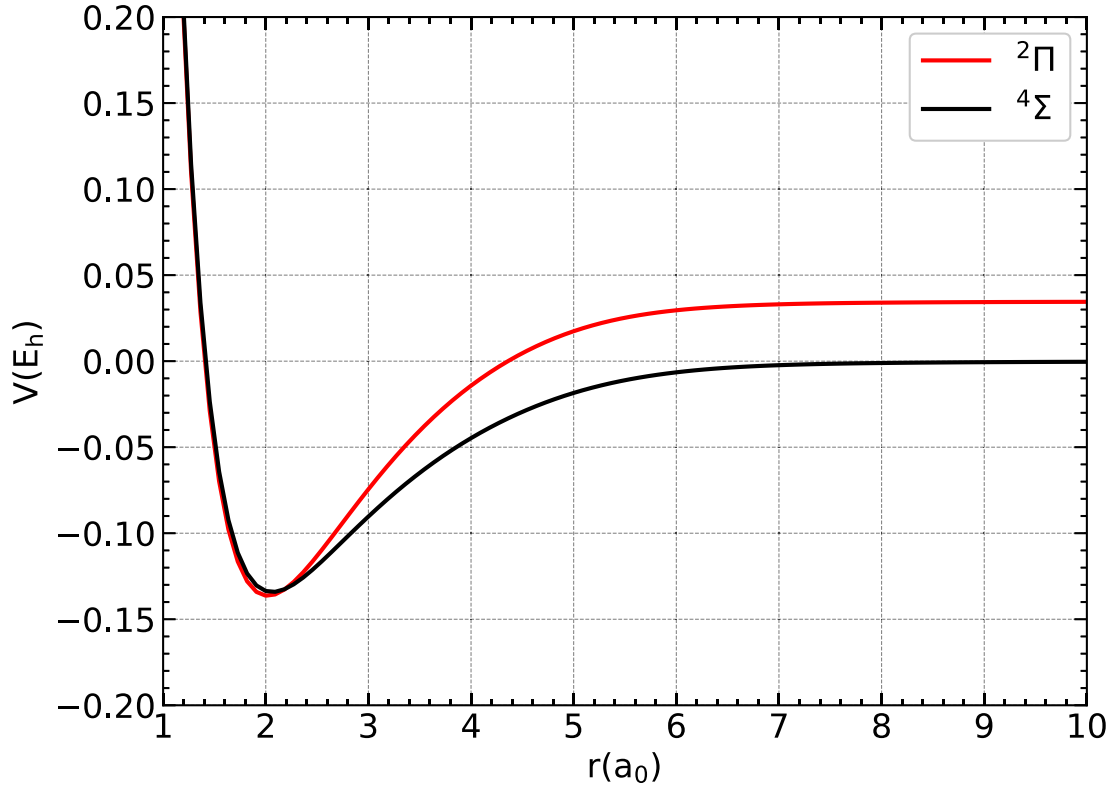


Fig. 1. Potential energy curves relevant for the spin-orbit induced $N(^4S) + H^+ \rightarrow N(^3P) + H(^2S)$ charge transfer.

2. Potential energy curves and spin-orbit coupling constant

Contemporary *ab initio* methods allow calculations of PECs at very high energies, even thousands of hartrees. At these energies, all the curves that describe the interaction of given atoms (in all excited states) merge, reaching the limit of the Coulomb interaction of nuclei [20].

Fig. 1 shows the $^4\Sigma$ and $^2\Pi$ PECs calculated in [20]. Charge transfer in $N(^4S) + H^+ \rightarrow N(^3P) + H(^2S)$ is described by the transition from the $^4\Sigma$ state (dissociating to $N(^4S) + H^+$) to $^2\Pi$ state (dissociating to $N(^3P) + H(^2S)$). The transition occurs, as assumed by the Landau-Zener theory, mainly at the crossing point, which is at $r = 2.1907$ bohr. At shorter distances, those curves remain very close to one another. That similarity of PECs is an uncommon and peculiar feature of that system.

The activation energy of excitation (the difference in electronic state energies in the dissociation limit) is 0.03471891 hartree. The activation energy calculated in this study agrees much better with the experimental value of 0.9467 eV (0.03479057 hartree) than the one calculated by Lin et al. [8].

The spin-orbit coupling constant V_{SO} was calculated from the full spin-orbit matrix (i.e., from the matrix elements for each spin multiplicity of both crossing states) calculated in MOLPRO according to [21]:

$$V_{SO} = \sqrt{\sum_{M,M'} |H_{SM,S'M'}^{SO}|^2}. \quad (1)$$

The present study is based on the coupling calculated at 0.001–15 bohr; at 15 bohr, it becomes zero. A significant increase of the spin-orbit coupling constant at short distances, which is shown in Fig. 2, was also observed in Ref. [22] for $^1\Sigma^+ - ^3\Pi$ curves of the ZnH^+ ion. At the curve crossing, the spin-orbit coupling constant is 46.70 cm^{-1} and remains the same or higher at short distances until 0.1 bohr (at shorter distances, it starts decreasing). The charge transfer in the reverse reaction of $N(^4S) + H^+$, occurring at the same crossing point, was studied by Butler and Dalgarno [23]. That study used the coupling of 50 cm^{-1} in the 1.5–3.5 bohr region.

The described PECs and spin-orbit coupling constant were calculated in MOLPRO software; the PECs were already calculated in Ref. [20]. For the present study, the spin-orbit coupling constant was additionally calculated. The curves were calculated with the MRCI method with the F12 explicit correction and the size extensivity Davidson correction to estimate the full configuration interaction. For hydrogen atoms, an aug-cc-pVQZ-F12 basis set was used, while for nitrogen atom, the core-valence correlation basis set aug-cc-pCVQZ-F12 was used; the spin-orbit matrix was calculated with the *hlsmat* option.

3. Cross section and rate constant methodology

The rate constant is given by

$$k(T) = \left(\frac{8}{\pi \mu (k_B T)^3} \right)^{1/2} \int_0^\infty \sigma_{CT}(E) E \exp(-E/(k_B T)) dE, \quad (2)$$

where the charge transfer cross-section is

$$\sigma_{CT}(E) = \int_0^{(1-E_x/E)r_x} 2\pi b P_{CT}(E, b) db, \quad (3)$$

where $P_{CT}(E, b)$ the charge transfer probability; r_x is the crossing position and E_x are the activation energy of the charge transfer process.

The probability P_{CT} within the Landau-Zener theory is given by

$$P_{CT}(E, b) = 2p(E, b)(1 - p(E, b)), \quad (4)$$

$$p(E, b) = \exp\left(-\pi \sqrt{\frac{2\mu}{E - b^2 E/r_x^2 - E_x}} \frac{V_X^2}{|F_1 - F_2|}\right). \quad (5)$$

The present calculations were performed both with the standard Landau-Zener method and with a combination of the Landau-Zener method and one-dimensional Gaussian wavepacket dynamics, which was applied at energies above 0.1 hartree. The Landau-Zener method was recently verified as a useful approach for various nonadiabatic transitions in atomic collisions involving nitrogen and oxygen atoms at temperatures reaching 50,000 K, including the spin-orbit coupling mechanism [17,24].

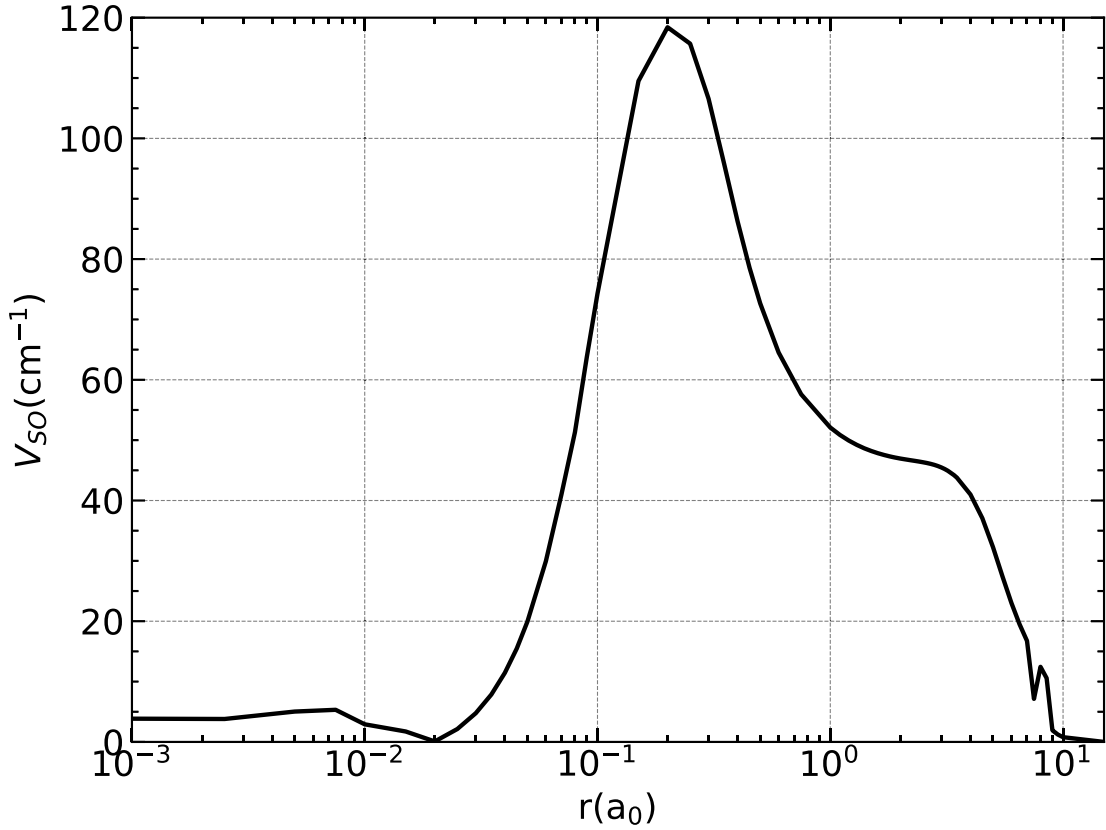


Fig. 2. Spin-orbit coupling constant of the $^4\Sigma$ and $^2\Pi$ PECs.

Because of the closeness of PECs and the magnitude of the spin-orbit coupling constant at distances smaller than the crossing point, high energies require a methodology that accounts for behavior at that short-range region. The energy and impact parameter are combined in the expression $E - b^2 E/r_x^2$, so that the probability is effectively a function of one variable $P_{CT}(E - b^2 E/r_x^2)$. The same relation is also used to transform the wavepacket dynamics probabilities $P_{CT}(E)$ calculated at $b = 0$ at various energies ($E \in [0.03472891, 100]$) to relate energy and non-zero impact parameter: $P_{CT}(E - b^2 E/r_x^2)$.

The procedure described above allowed employing charge transfer probabilities from one-dimensional ($b = 0$) wavepacket dynamics, which was performed with the numerically exact quantum dynamics with the split-operator method, and also compared with discrete variable representation as implemented in Ref. [25].

The split-operator method employed a grid ranging from 0.00001 to 32 bohr, discretized into 16384 points. The time step was selected according to the initial kinetic energy of the wavefunction: 1 a.u. was used for $E_{kin} < 10$, and 0.1 a.u. for $E_{kin} > 10$. Charge transfer probabilities were evaluated when the wavepacket reflected from the nitrogen atom and returned to its initial position, ensuring minimal interaction with the grid boundaries. All simulations were verified for convergence with respect to both grid size and time step.

The Gaussian wavepackets (of width 1 bohr) were evolved from an interatomic distance of 15 bohr and back (see Fig. 1) in both approaches. The nonadiabatic evolution of wavepacket placed at the lower state yields the density matrix, the diagonal elements of which are the probabilities $P(E)$ of each state after the scattering event; the resulting probability of the higher state is the charge transfer probability $P_{CT}(E, b) = P_{CT}(E - b^2 E/r_x^2)$.

At energies lower than 0.1 hartree, the Landau-Zener method was used, and at higher energies, transition probabilities were calculated with Gaussian wavepacket dynamics. That choice of energy was made because at energies close to 0.1 hartree, transition probabilities of the Landau-Zener and wavepacket dynamics are similar.

The scaling formula for the cross section from the hydrogen case to deuterium values, recasted from velocity to energy, is given by [15]

$$\sigma_D(E) = \frac{1 - V(r_x)/E}{1 - V(r_x)/(E\mu_H/\mu_D)} \sigma_H(E\mu_H/\mu_D), \quad (6)$$

where V is the initial state PEC, and μ 's are the reduced masses. The cross sections were tested according to that scaling.

All cross sections and rate constants reported in this study should be multiplied by the effective electronic degeneracy, which in this case is $g^{SOC} = 1/2$ [26]. The results are reported without that factor because the values against which they are compared do not explicitly state either the inclusion of such a degeneracy or the value of other multipliers used.

4. Results

For comparison, the cross section and the rate constant were first calculated solely with the Landau-Zener method. Fig. 3 shows the cross sections for all isotopes studied and the deuterium values scaled from hydrogen data according to Eq. (6). It allowed to conclude that the isotope effect is significant in cross section and that the scaling is inadequate at the most crucial energies used in the calculation of rate constants.

The rate constants, given in Fig. 4, show that the isotope effect is negligible. The increase in cross section for heavier isotopes is largely compensated by increased mass in the denominator of Eq. (2). Furthermore, the rate reported by Kingdon and Ferland [7] is not correct even qualitatively, only the order of magnitude is correct; it can be explained by the fact that it was estimated from the reverse reaction, which, as barrierless, is not correctly described by the Landau-Zener theory [17].

Figs. 5, 6, 7 present the cross sections of charge transfer involving hydrogen, deuterium, and tritium in the wavepacket methodology compared with the results of the Landau-Zener method. The deviation from the Landau-Zener values is the most significant at high energies and

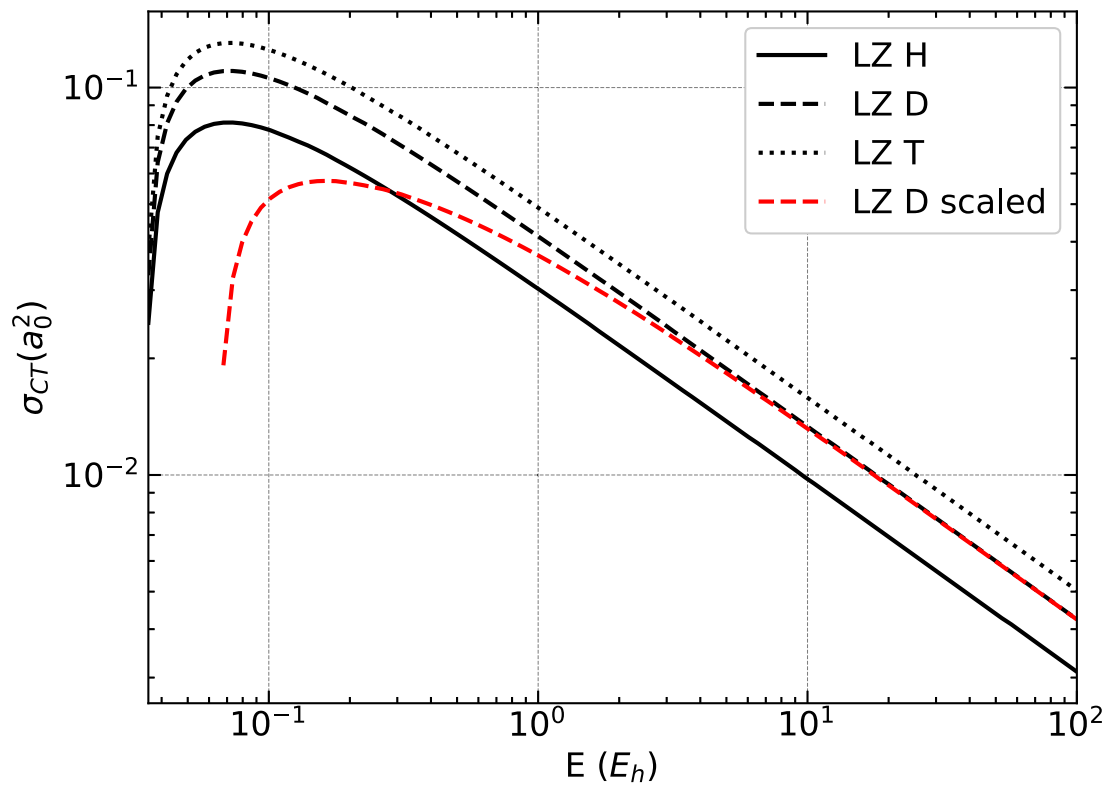


Fig. 3. The Landau–Zener charge transfer cross sections for reactions involving hydrogen (LZ H), deuterium (LZ D), and tritium (LZ T). The result of scaling the hydrogen cross section to deuterium (LZ D scaled) is also given.

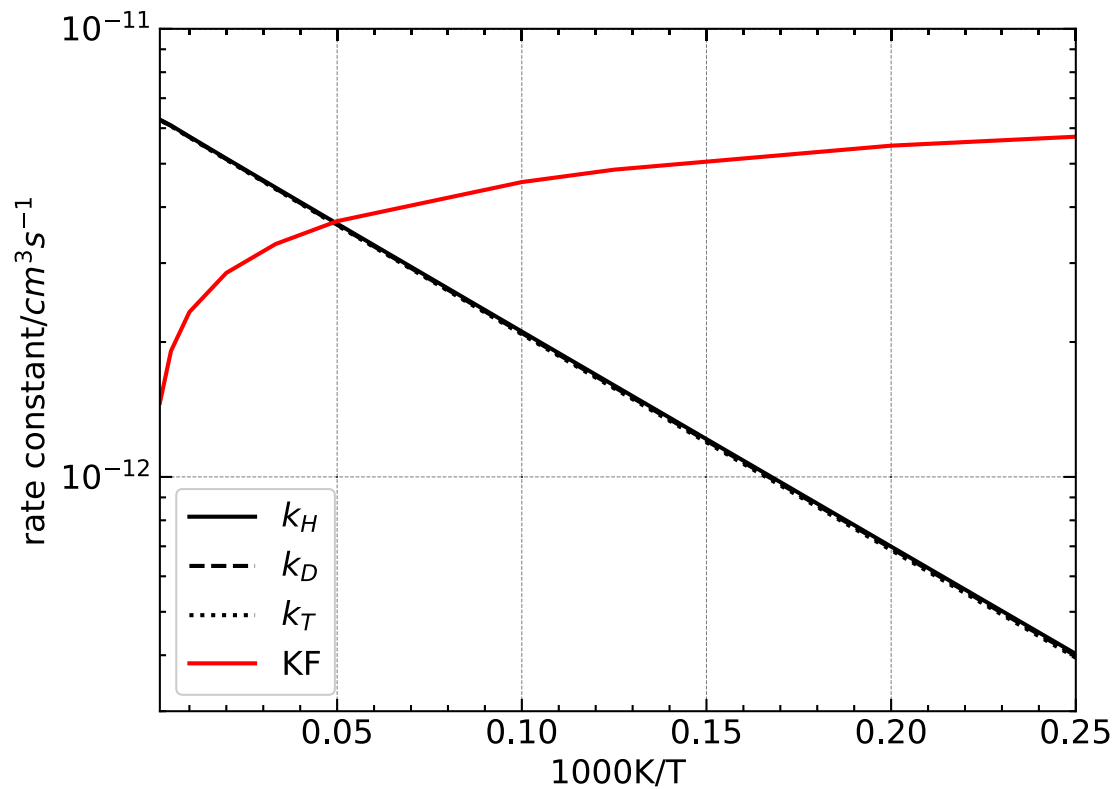


Fig. 4. The Landau–Zener charge transfer rate constant at temperatures 4000 K–50,000 K for reactions involving hydrogen (k_H), deuterium (k_H), and tritium (k_H). The rate given in Kingdon and Ferland [7] work is shown (KF).

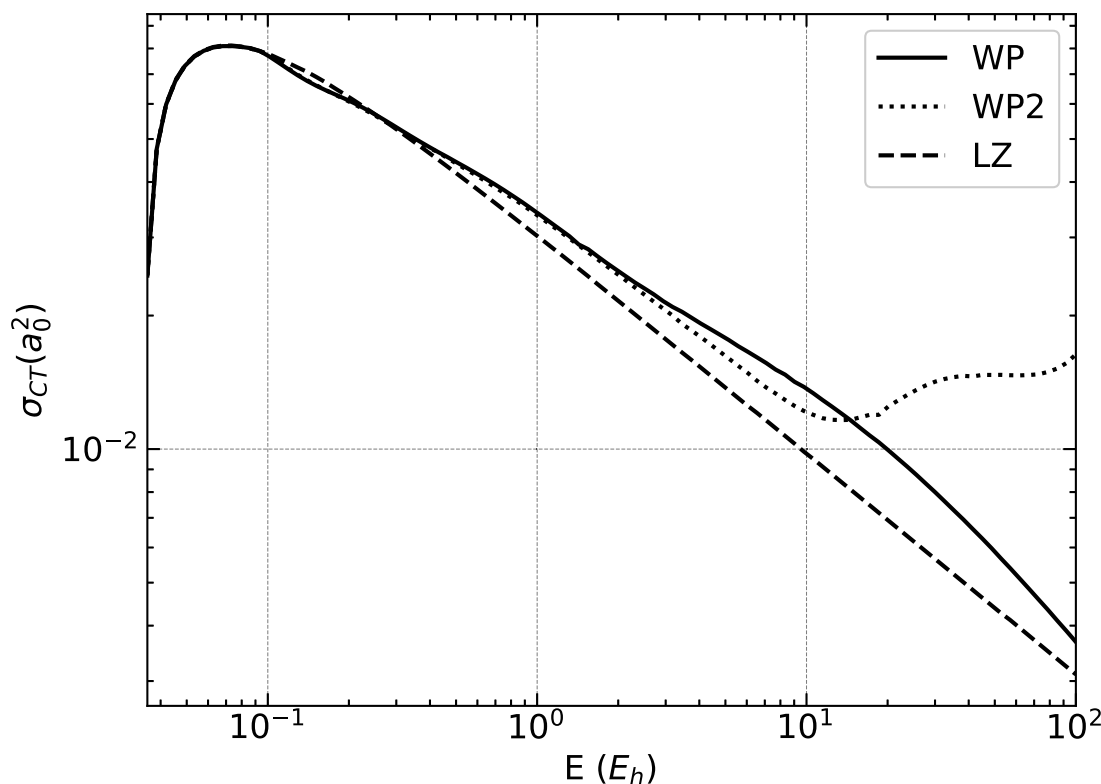


Fig. 5. The cross sections of charge transfer involving hydrogen in the wavepacket split-operator methodology (WP) and the wavepacket discrete variable representation methodology (WP2) [25] compared to the Landau–Zener method.

increases with isotope mass. Fig. 5 compares both the split-operator code and the discrete variable representation code. The former one features a sudden increase in the cross section, which most probably is incorrect because the coupling decreases at the highest energies.

To estimate the influence of the short-range portion of interaction, in Fig. 8 the wavepacket cross section based on full coupling was compared to the dynamics with coupling that is zero below 0.5 bohr - it was found that the cut-off coupling values are smaller (and closer to the Landau–Zener cross sections) at high energies, i.e. the coupling at the short interatomic distances is non-negligible and influences the cross section at energies over 1 hartree, as was expected.

Inspecting the cross sections, it can be predicted that deviations from the Landau–Zener method in the rate constant will be significant only at very high temperatures. Fig. 9 shows that the deviation from the straight-line Arrhenius behavior of the Landau–Zener method is increasingly relevant above 50,000 K for hydrogen. For heavier isotopes, deviation from the straight line increases with isotope mass and is relevant at lower temperatures (starting at around 20,000 K); see Fig. 10, which is focused on the highest temperatures.

Lin et al. [8] discussed the relative importance of the spin–orbit coupling mechanism. In contrast to the conclusion based on the previous spin–orbit rates that this mechanism is crucial below 20,000 K, the spin–orbit mechanism is not negligible below 60,000 K and dominates the overall rate below 30,000 K (by comparison of the red dots in Fig. 9).

5. Conclusions

The charge transfer in the spin–orbit induced process $\text{N}(^4\text{S}) + \text{H}^+ \rightarrow \text{N}^+(\text{}^3\text{P}) + \text{H}(^2\text{S})$ and its isotope analogues was studied. The rate constants are reported and shown to be qualitatively different from the older values based on the spin–orbit coupling mechanism, which used approximate potentials and an indirect approach. The comparison with

rates based on radial and rotational coupling showed that the spin–orbit coupling mechanism remains important at 30,000 K–60,000 K, in contrast to previously assessed 20,000 K.

The isotope effect was found to be important in the case of the rate constant only at the highest temperatures, but its effect is significant in cross sections. The scaling proposed for the isotope effect is incorrect below 1 hartree, and its range of usefulness is limited to high energies.

The combination of the Landau–Zener theory with a methodology based on one-dimensional Gaussian wavepacket propagation allowed us to take into account the increase of cross section and rate constant at the higher energies and temperatures, which is present due to transition probability beyond the crossing point.

CRediT authorship contribution statement

M. Buchowiecki: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **T. Jřra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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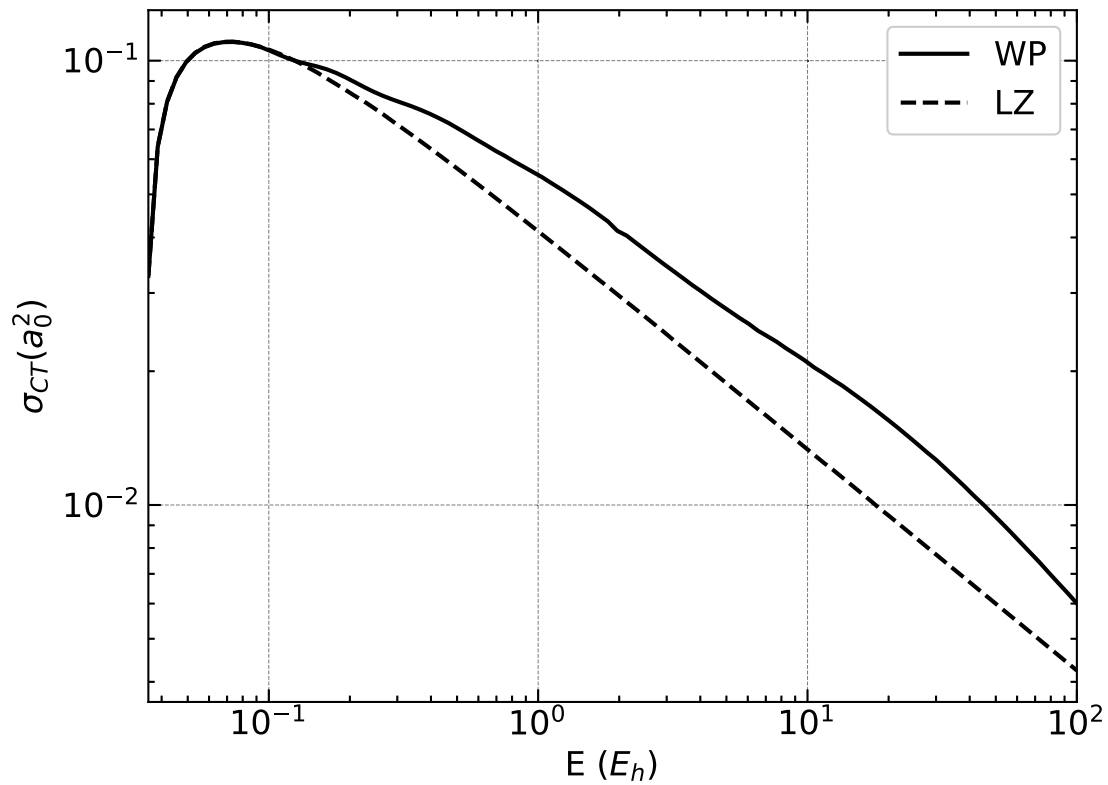


Fig. 6. The cross sections of charge transfer involving deuterium in the wavepacket split-operator methodology (WP) compared to the Landau-Zener method.

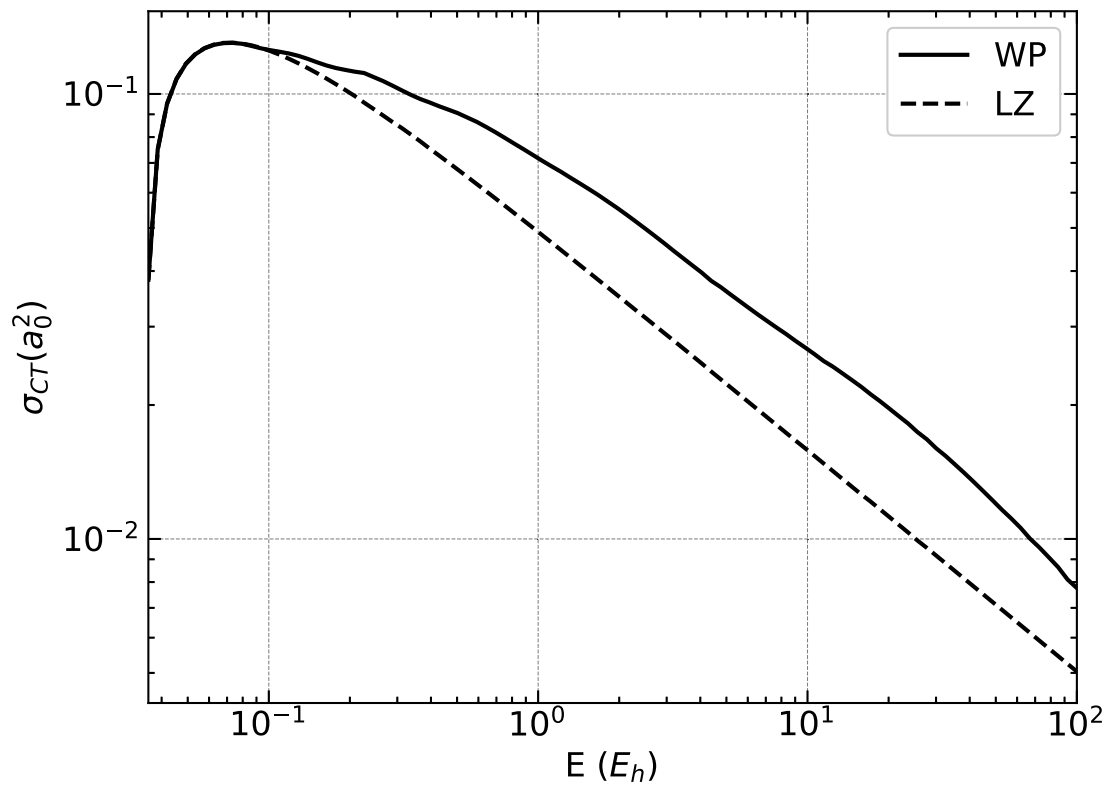


Fig. 7. The cross sections of charge transfer involving tritium in the wavepacket split-operator methodology (WP) compared to the Landau-Zener method.

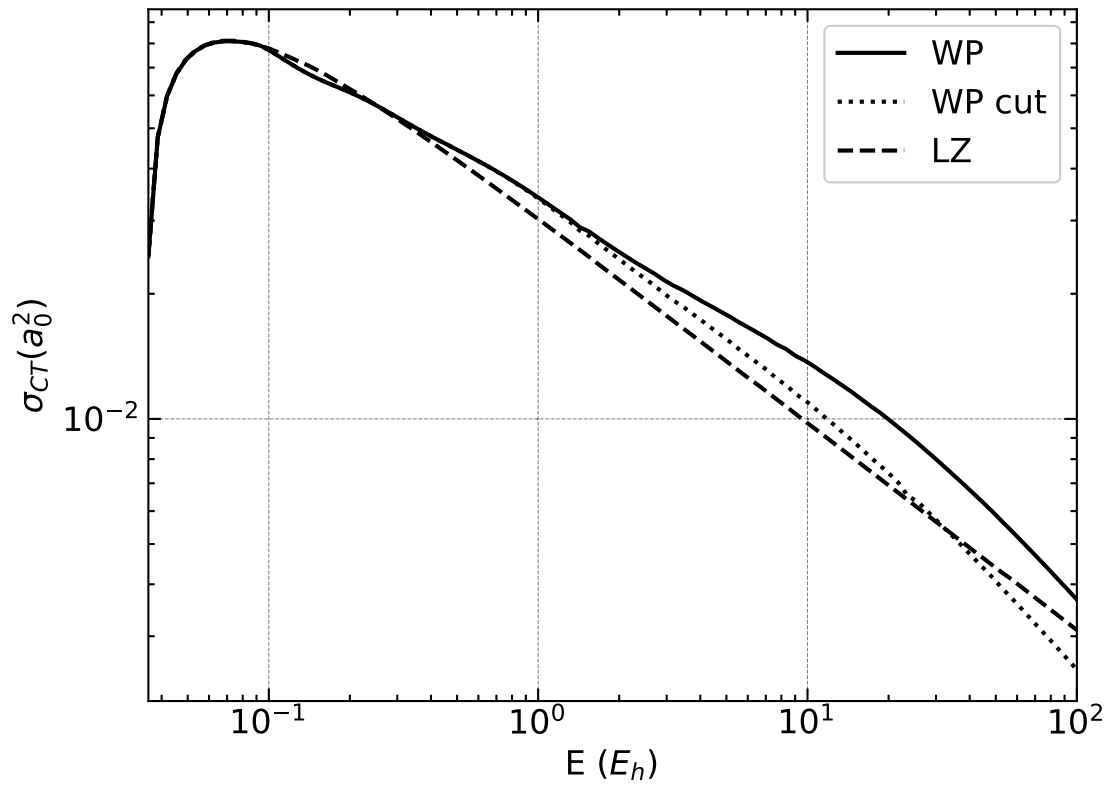


Fig. 8. The cross sections of charge transfer involving hydrogen in the wavepacket split-operator methodology (WP) compared with the results with the coupling set to zero below 0.5 bohr (WP cut).

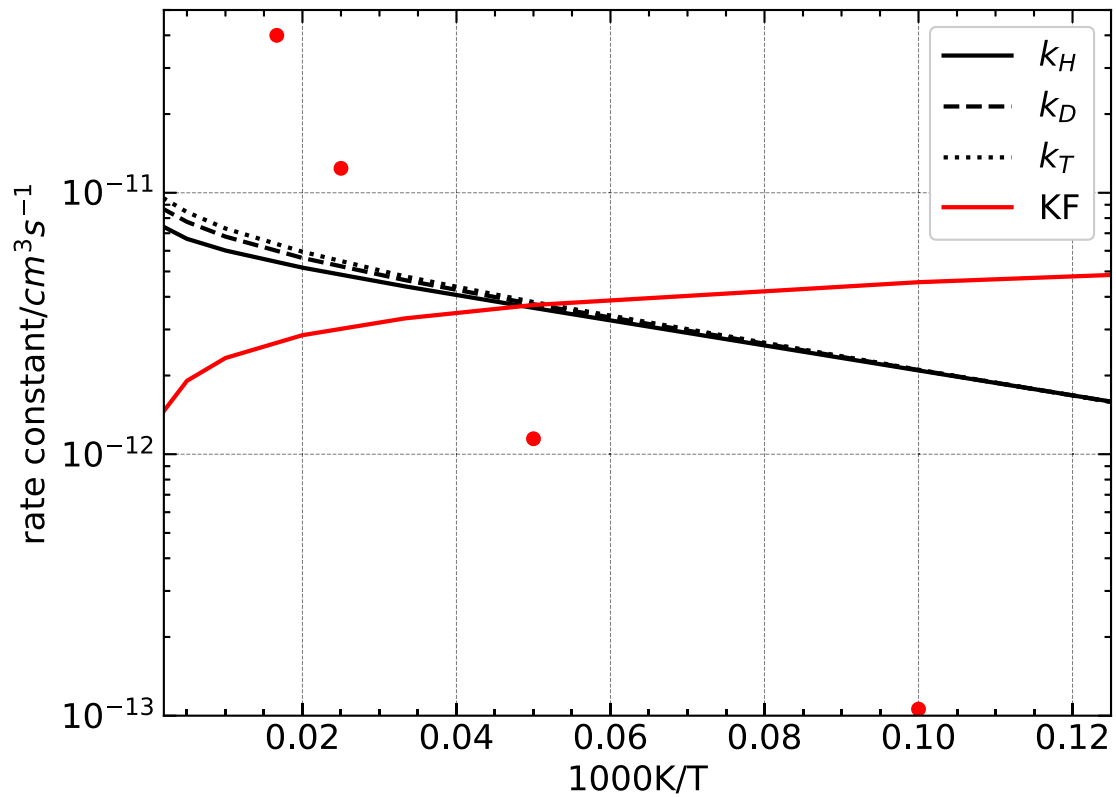


Fig. 9. The rate constants at temperatures 8000 K–500,000 K for reactions involving hydrogen (k_H), deuterium (k_H), and tritium (k_H) in wavepacket methodology. The rate given in Kingdon and Ferland [7] work is shown (KF). Red dots show the Lin et al. values [8] based on radial and rotational coupling.

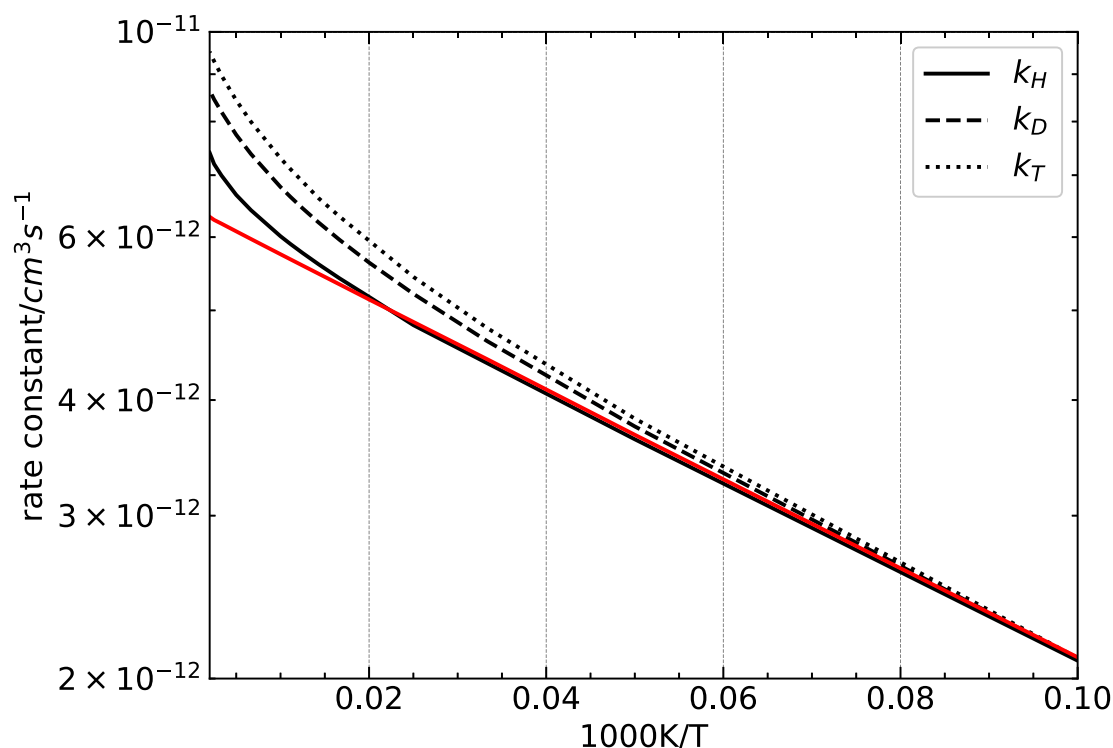


Fig. 10. The rate constants at temperatures 8000 K–500,000 K for reactions involving hydrogen (k_H), deuterium (k_D), and tritium (k_T) in wavepacket methodology. The straight red line for the hydrogen Landau–Zener case is shown for assessment of deviations.

Appendix A

In Fig. 11, the methodology proposed in this study was compared with the method based on two-dimensional wavepackets that have been evolved at impact parameters in the range from zero to 5 or 10 bohr, explicitly yielding the $P(b, E)$ function. The cross sections were calculated as:

$$\sigma_{CT}(E) = \int_0^\infty 2\pi b P_{CT}(E, b) db, \quad (7)$$

the $b = 5$ as the upper bound of the integral was sufficient to converge the results, except for the lowest energy point, where 10 bohr was used. Additionally, the quantum scattering results (based on solving radial coupled equations) confirm values from two-dimensional wavepacket dynamics, except for the lowest energies.

The results of the two-dimensional approach are found to oscillate around the Landau–Zener values. In particular, all methods confirm an increase in the cross section (with respect to the Landau–Zener method) at energies around 10 hartree. However, at lower temperatures, the methodology described in this study underestimates the exact cross section.

Despite the underestimation of the cross sections at low energies, the methodology proposed in this research is expected to be useful for studying processes with typical crossing geometry also at lower energies; the present case has an untypical similarity of the PECs involved in the process (even in the vicinity of the crossing point), which causes the cross section to be significantly higher at lower energies than the one from the Landau–Zener method. Note that the Landau–Zener method was shown to successfully estimate the rate constant of a process with the typical geometry of the crossing region [17].

Preliminary results of three-dimensional wavepacket dynamics at the energy of 0.04 hartree, show that the agreement with quantum scattering is better. Most probably, a much finer grid is needed for such wavepacket evolution, and that makes such an approach extremely costly.

Table A.1

Cross sections from two-dimensional wavepacket dynamics (red dots on Fig. 11).

E	$\sigma_{CT}(E)$
0.06	0.20467
0.1	0.14007
0.2	0.08411
0.4	0.04716
0.7	0.03392
1.0	0.02513
2.0	0.01766
4.0	0.01442
6.0	0.01410
10.0	0.01288
20.0	0.005597
50.0	0.003541

Finally, agreement with the quantum scattering method at a broad range of energies shows that there is no need for computationally very expensive three-dimensional wavepacket dynamics, except for the lowest energies (i.e., just above the activation threshold). The suggested methodology for such calculations at a broad energy range can be a combination of close-coupling quantum scattering at the lowest energies (the region where such calculations are the least demanding) with two-dimensional wavepacket dynamics at high energies (the region where very expensive three-dimensional wavepacket dynamics is not needed).

Appendix B. Supplementary data

The charge transfer cross sections for $H^+/D^+/T^+-N(^4S)$ collisions are given in sigma.txt. The spin–orbit coupling constants are given in SOCC.txt

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jqsrt.2025.109697>.

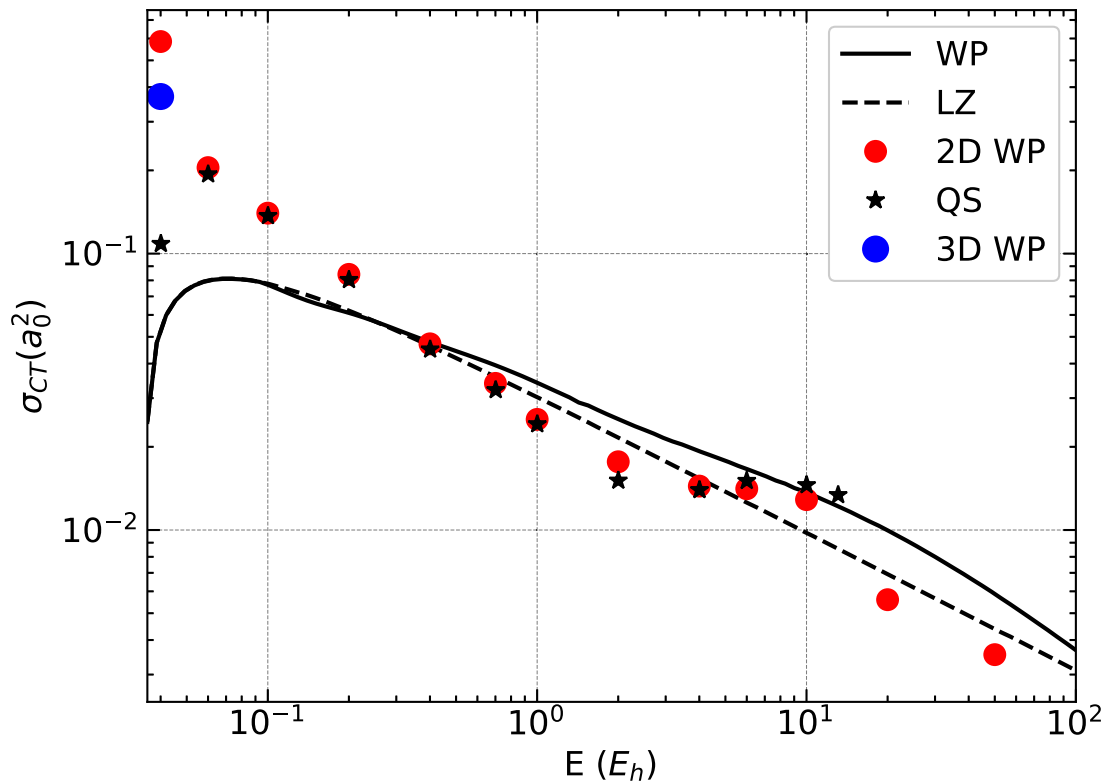


Fig. 11. The cross sections of charge transfer involving hydrogen in the wavepacket split-operator methodology (WP) and the Landau–Zener method (LZ) compared with two-dimensional wavepacket results (2D WP) and quantum scattering results (QS). Approximate value calculated with three-dimensional wavepacket dynamics at $E = 0.04$ is given as a blue dot (3D WP).

Table A.2

The recommended rate constants calculated from two-dimensional wavepacket cross section data with the correction from quantum scattering at the lowest energies.

T	Rate constant
4000	$9.059 \cdot 10^{-13}$
5000	$1.569 \cdot 10^{-12}$
10 000	$4.370 \cdot 10^{-12}$
20 000	$6.514 \cdot 10^{-12}$
30 000	$7.045 \cdot 10^{-12}$
50 000	$7.103 \cdot 10^{-12}$
100 000	$6.567 \cdot 10^{-12}$

Data availability

Data will be made available on request.

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